Foundations of Deep Learning

List of topics for the written questions
- Theory
- Practical part

F.A.Q.s
- General
- Written questions
- Project and presentation

List of topics for the written questions

Theory

L01 - Introduction to Deep Learning:
- Definition of Deep Learning
- Machine vs Deep Learning
- Data-centric AI vs Model-centric AI
- Feature extraction in machine learning and deep learning
- Enabling factors that made Deep Learning possible

L02 - Linear Algebra:
- Eigendecomposition procedure
- Singular value decomposition
- Principal Component Analysis

L03 - Machine Learning basics:
- Convex and nonconvex optimization
- Gradient Descent algorithm and its variants
- Empirical risk, underfitting, and overfitting
- Regularization

L04 - Feed Forward Networks
- Perceptron Neurons
- Multi-layer Perceptron Networks
- Activation functions
- Backpropagation

Practical part

E01 - Principal Component Analysis:
- Code understanding
- Shape (dimensionality) of the variables involved in the transformation
- Usage of explained variance

E02 - Optimizers:

- Code understanding
- Usage of callbacks

E03 - Regression from tabular data:
- Code understanding
- Usage of Keras commands (dataset, compile, fit, evaluate, predict)
- Properties of layers (trainability, dimensionality of input/parameters/output)

E04:
- Code understanding
- Usage of Keras commands (dataset, compile, fit, evaluate, predict)
- Properties of layers (trainability, dimensionality of input/parameters/output)
- Usage of callbacks for early stopping

F.A.Q.s

General

Q: How is the exam structured?
A:
1. Written questions.
2. Group project + presentation.

For every exam date, we will organize both a written exam and a project presentation exam.
The written question part has a special additional date in April.

Q: Is there an expiration date for the various parts?
A:
The student can repeat any of the individual parts (written questions, or project+presentation).
The student can take each part on a different date; however, the two parts must be completed within the same session:

(April)-June-July-September is the summer session
January-February is the winter session

Q: How are the parts weighted?
The student will receive a global grade, determined based on:
⅓: Written questions
⅓: Project (code and formal correctness)
⅓: Presentation (quality of exposition and answers to individual questions)
Different members of a single group might therefore get a different grade.

Written questions

Q: Will the exam be taken on our computer?
A:

No, the written questions are on paper.
Bring your pen.

Q: What language should we use for our answers?
A:
English

Q: How many questions will the exam have?
A:
A total of 6 questions:
3 from the theory (sampled from lectures L01 to L04).
3 from the practical part (sampled from the laboratory exercises E01 to E04).

Q: Are there going to be linear algebra exercises?
A:
No.

Q: What types of questions are in the practical part?
A:
They are either checkbox answers, or you will be required to specify the dimensionality of variables involved in a snippet of code.
You will not be asked to write code in the written part of the exam.

Q: How long does the written exam take?
A:
1 hour and 15 minutes.

Project and presentation

Q: In what language will we present our project?
A:
English, with slides written in English.

Q: When will the projects be assigned?
A:
After the April partial exam, you will be given more information about the possible project.
Datasets are made available by the teacher, but the students can **propose** external datasets as well.

Q: What is the number of people per group?
A:
2 or 3 per group.
You will be able to autonomously organize in groups via e-learning.
We ask you to provide this information as soon as possible, so as to facilitate the organization of presentations.

Q: How to submit the project material?
A:

For each group, a representative must submit the project via eLearning.
The submission must include:

- Slides in PDF format that will be used for the presentation.
- Developed Python code.
- Trained models (the weights saved from Keras, for example, in .h5 or .keras format) (due to their size, trained models can be a link to a Google Drive folder, to be provided within the eLearning submission).

DO NOT submit any material or link via email.
DO NOT submit slides and code as Google Drive links (only trained models can be shared as external links in case of size limitations).

Q: When is the project to be submitted?
A:
The code and the presentation must be submitted one week before the official exam date. (23.59 p.m. Italy time on the same weekday, one week prior).
Small modifications on your end are also allowed during the last week, but you must match the deadline via eLearning submission.

Q. When is the date for the project presentation?
A:
The official date for each exam refers to both written questions and project presentations.
Based on the number of registered students, the groups might be split into 2 days with respect to the official exam date.
You will receive an early communication of the split.

Q: How long does the presentation take?
A:
Each member of the group must present for 5 minutes.
Additional time will be allocated for individual questions.

Q: How to structure the presentation (and the project)
A:

1. Introduce yourself! (list the group members)
2. Introduce the problem you are trying to solve
3. Show an analysis of the available data
4. Describe your solution
5. Comment on the positive and negative results of your solution (rely on quantitative and qualitative observations)
6. Describe alternative (possibly failed) experiments

Points 4,5,6 can be rearranged at will.
Consider 1 minute ≈ 1 slide (e.g. 3 students = 15 minutes = 15 slides)

Q: Can we use pretrained models?
A:
Yes, but the project cannot rely only on pretrained models.
Examples of allowed applications of a pretrained model are:

1. Using a pretrained model for feature extraction / embedding, on top of which your own neural layers / models must be trained.

2. Comparing your own “trained from scratch” solution to an existing one.
3. Transfer learning from an existing model to a different task.

Q: I want to use a certain dataset but it is too large. Can I subsample it?
A:
Yes, if the original dataset is too large, you can subsample it.
Be sure to address this explicitly during the presentation.
In general, you want to aim for a global random sampling.
If the problem is too complex (e.g. 10000 classes), you might also consider sampling by removing certain classes.
Removing classes to make the task easier is generally discouraged, and it should be well-motivated.